J Biol Inorg Chem (2015) 20:905–919
DOI 10.1007/s00775-015-1275-1

ORIGINAL PAPER

Thermodynamics of Calcium binding to the Calmodulin N-terminal domain to evaluate site-specific affinity constants and cooperativity

Maria Rosa Beccia[1,2,3] · Sandrine Sauge-Merle[1,2,3] · David Lemaire[1,2,3] · Nicolas Brémond[1,2,3] · Romain Pardoux[1,2,3] · Stéphanie Blangy[4,5] · Philippe Guilbaud[6] · Catherine Berthomieu[1,2,3]

Received: 22 February 2015 / Accepted: 31 May 2015 / Published online: 13 June 2015
© SBIC 2015

Abstract Calmodulin (CaM) is an essential Ca(II)-dependent regulator of cell physiology. To understand its interaction with Ca(II) at a molecular level, it is essential to examine Ca(II) binding at each site of the protein, even if it is challenging to estimate the site-specific binding properties of the interdependent CaM-binding sites. In this study, we evaluated the site-specific Ca(II)-binding affinity of sites I and II of the N-terminal domain by combining site-directed mutagenesis and spectrofluorimetry. The mutations had very low impact on the protein structure and stability. We used these binding constants to evaluate the inter-site cooperativity energy and compared it with its lower limit value usually reported in the literature. We found that site I affinity for Ca(II) was 1.5 times that of site II and that cooperativity induced an approximately tenfold higher affinity for the second Ca(II)-binding event, as compared to the first one. We further showed that insertion of a tryptophan at position 7 of site II binding loop significantly increased site II affinity for Ca(II) and the intra-domain cooperativity. ΔH and ΔS parameters were studied by isothermal titration calorimetry for Ca(II) binding to site I, site II and to the entire N-terminal domain. They showed that calcium binding is mainly entropy driven for the first and second binding events. These findings provide molecular information on the structure–affinity relationship of the individual sites of the CaM N-terminal domain and new perspectives for the optimization of metal ion binding by mutating the EF-hand loops sequences.

Keywords Calmodulin · Calcium · Microscopic binding constants · Thermodynamics · Cooperativity

Introduction

The regulation of protein function often depends on how they interact with metal ions or organic molecules. Many proteins involved in cell signaling or metabolic pathways contain multiple binding sites and interact simultaneously with different ligands. In these multisite proteins, the binding of a given ligand may induce changes in the protein conformation that favor or impair the binding of other ligands at other sites of the protein [1, 2]. This mechanism, known as allostery, plays a pivotal role in cell regulation processes, as it allows communication between protein subunits at a distance. Moreover, it enhances protein sensitivity and enables a fast response to external stimuli. This is achieved by producing great variations of protein output in response to small changes in ligand concentration [3]. Therefore, understanding the molecular factors and mechanisms that govern allosteric binding processes is of great

Electronic supplementary material The online version of this article (doi:10.1007/s00775-015-1275-1) contains supplementary material, which is available to authorized users.

✉ Catherine Berthomieu
catherine.berthomieu@cea.fr

1 CEA, DSV, IBEB, Laboratoire des Interactions Protéine Métal, Saint-Paul-Lez-Durance, France

2 CNRS, UMR Biologie Végétale et Microbiologie Environnementale, Saint-Paul-Lez-Durance, France

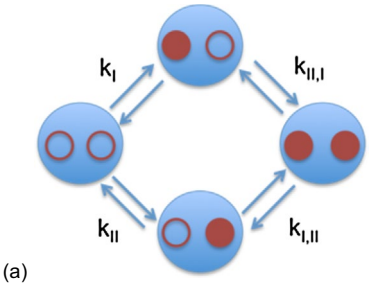
3 Aix-Marseille Université, BVME, 13284 Marseille, France

4 Aix-Marseille Université, Architecture et Fonction des Macromolécules Biologiques, 13284 Marseille Cedex 09, France

5 CNRS, UMR7257, Architecture et Fonction des Macromolécules Biologiques, 13288 Marseille Cedex 09, France

6 CEA, DRCPC, SMCS, LILA, Bagnols Sur Cèze, France

importance, and remains one of the main challenges in biochemistry. This requires the characterization of the apo- and holo-protein, as well as the intermediate microstates. For instance, in a two-site protein, four different microstates are achievable and the relative population of each state depends on the microscopic thermodynamic parameters of the system (Scheme 1). Evaluating these microscopic thermodynamic parameters is not always straightforward for multisite proteins, especially when cooperative processes are involved. Only macroscopic binding constants can be measured in experimental situations, which are limited to reporting on the total number of half-liganded and totally liganded states.



Scheme 1 **a** Schematic diagram of the Ca(II)-binding pathways in the CaM N-terminal domain. *Solid red circles* indicate that calcium is present in the site. Microscopic binding constants are indicated for each binding step. **b** Macroscopic representation of the reactions, corresponding to Eqs. (1) and (4) from the Results section. **c** Equations describing the relationship between the macroscopic binding constants, K_1 and K_2, and the microscopic binding constants k_I, k_{II}, $k_{I,II}$ and $k_{II,I}$

In this context, the EF-hand protein calmodulin (CaM) is a fascinating model system that has been largely studied over the past decades, due to its versatility and ubiquity in eukaryotic cells [4–10]. By cooperatively binding up to four calcium cations, CaM is activated to modulate more than 100 target proteins involved in numerous physiological processes including cell proliferation or programmed cell death [11–13, 14]. The crystal structure of Ca(II)-saturated calmodulin indicated that CaM is arranged into two similar globular domains separated by a central helix [15, 16]. Each domain consists of a pair of EF-hand helix-loop-helix Ca(II)-binding motifs [17, 18], commonly referred to as site I, site II (N-terminal domain), site III or site IV (C-terminal domain). The Ca(II)-binding loops consist of a 12-amino-acid sequence, six of which provide Ca(II) ligands (at positions 1, 3, 5, 7, 9 and 12 of the loop) arranged in a pentagonal bipyramidal geometry (Fig. 1) [16, 19]. Ca(II) binding induces conformational changes, which result in an intra-domain cooperative effect [20]. It leads to changes in α-helix orientation and in the exposure of hydrophobic residues to the solvent [21, 22]. This domain opening is an important factor determining the Ca^{2+}-binding properties of CaM [23]. Two-dimensional NMR experiments showed that the N- and C-terminal domains have different conformational dynamics in solution, with a stable closed

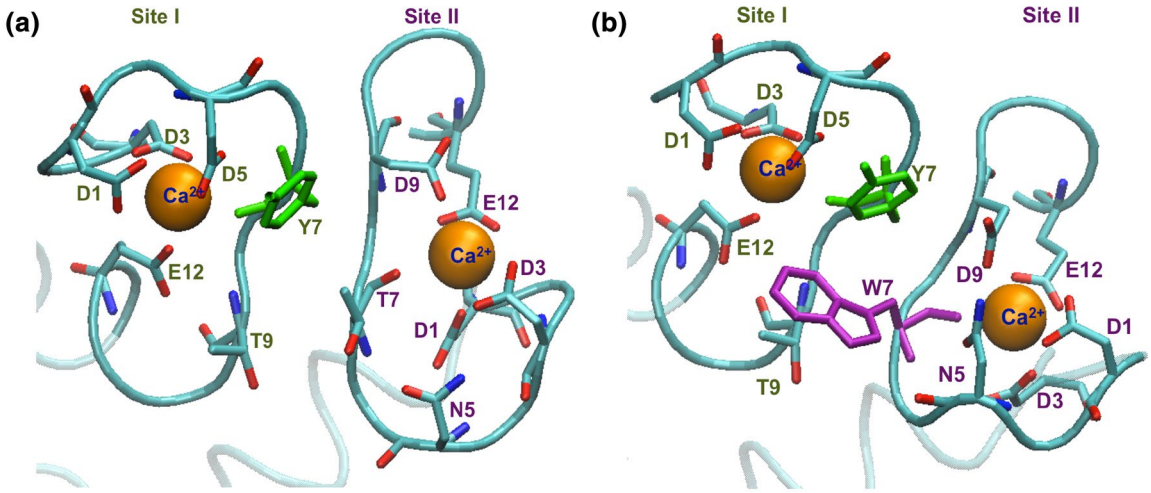


Fig. 1 Structural models of the Ca(II) binding loops I and II for the peptides analyzed in the present study **a** CaM Y I-II and **b** CaM YW I-II. Amino acids are numbered according to their position in the 12 residues Ca(II) binding loops. *D* stands for aspartate, *E* for glutamate, *Y* for tyrosine, *T* for threonine, *W* for tryptophan, *N* for asparagine. Calcium is coordinated by bidentate E12, monodentate D1, D3, and D5 in site I, (or by the carbonyl of N5 in site II), by the main chain carbonyl at position 7 and by a bridged water stabilized by the residue in position 9 (T for site I or D for site II). The models were obtained using molecular dynamic simulations, as detailed in Supporting Information S3

conformation of the apo-N-terminal domain in contrast to a more flexible C-terminal domain in the absence of calcium [21, 22, 24, 25]. A high degree of mobility near the middle of the central helix between the N- and C-terminal domains was also evidenced in solution [24, 26, 27].

A wide variety of methods have been used to study Ca(II) binding to CaM (reviewed in [28]). These studies showed that the two CaM domains have significantly different stabilities and macroscopic affinities for calcium despite similarities in their sequences and structures [21]: the C-terminal domain has a six to tenfold higher affinity than the N-terminal domain [29–31]. However, despite the many published reports on this system, quantitative studies regarding site-specific metal-binding affinities and inter-site cooperativity within one of the two domains of CaM have been limited [20, 32–34]. Lower limit values for inter-site cooperativity have been provided in the literature for the N-terminal and C-terminal domains of calmodulin, with the assumption that sites I and II or sites III and IV have identical specific calcium-binding properties [20, 29, 32, 35]. This is due to the difficulty in obtaining an accurate estimate of at least two of the four microscopic constants in Scheme 1, in addition to the macroscopic binding constants K_1 and K_2, which are necessary to evaluate the cooperative effect.

To the best of our knowledge, the only existing evaluation of site-specific microscopic constants of CaM was made by grafting the individual Ca(II)-binding loops into the domain 1 of the cell adhesion protein CD2 [34]. Inter-site cooperativity values deduced from this approach were by far larger than the lower limit estimates. However, this method aimed at determining intrinsic affinities of the Ca(II)-binding loop of each EF-hand site without taking into account the influence of flanking regions and conformational changes associated with metal binding and did not allow the evaluation of the microscopic constants for a partially metal-occupied N-terminal domain peptide ($k_{I,II}$ and $k_{II,I}$ of Scheme 1).

Another approach to estimate microscopic constants within a domain of calmodulin is the use of site-directed mutagenesis to alter or impair calcium binding at one of the two binding sites. Mutants targeting the Glu at position 12 of the Ca(II)-binding loop of site III or site IV were performed towards this end [36]. However, these mutations significantly affected the structural and dynamical properties of the protein variants, impairing the analysis of the binding cooperativity [36]. Indeed Glu at position 12 of the Ca(II)-binding loop was shown as a key determinant of the structural changes accompanying Ca(II) binding in EF-hand motifs [37–39]. Actually, amino acids of the C-terminal part of the Ca(II)-binding loop were shown to be highly structured in the apo-enzyme and residues at positions 7 and 8 of the loop are involved in an inter-site antiparallel beta sheet motif, important for domain stability and cooperative Ca(II) binding [40–42]. In contrast, amino acid residues of the N-terminal part of the loop were reported to be poorly structured and dynamic [23, 43] and references therein.

In this study, we have combined site-directed mutagenesis and a thermodynamic study of the Ca(II)-CaM N-terminal domain, to evaluate the site-specific Ca(II)-binding constants and hence the four calcium-binding microscopic constants in solution. The N-terminal domain was chosen because, in contrast to the C-terminal domain, it exhibits a closed compact conformation in the apo-form [21, 22, 24], limiting the impact of protein conformational dynamics to the structural changes associated with Ca(II)-binding events. In addition, we targeted residues of the N-terminal part of the Ca(II)-binding loops to impair Ca(II) binding at one or the other site, to minimize mutational effects on the protein structure and stability, as controlled by thermal unfolding measurements.

Results for peptides that are able to chelate Ca(II) at sites I and II (referred to as *two-site peptides*) and results obtained with peptides in which metal binding at site I or at site II was impaired (*one-site peptides*) allowed us to evaluate site-specific Ca(II)-binding affinity for each EF-hand motif, the enthalpy and entropy of Ca(II) binding, and to compare the calculated inter-site cooperativity energy values with the lower limit value ($\Delta\Delta G_{lim}$) usually reported in the literature. In addition, we investigated the effect of introducing a Thr64Trp mutation at position 7 of the Ca(II)-binding loop of site II, involved in the inter-site β-sheet structure, on the specific Ca(II) affinity of site II and on cooperativity (Fig. 1b, see Table 1 for the peptide sequences). Our approach provides quantitative information on the affinity of each binding site within the isolated N-terminal domain of calmodulin and its inter-site cooperativity. The basic structural and functional unit of several EF-hand proteins is a pair of EF-hand motifs that form a stable domain, whose activity is often linked to an inter-site cooperative effect. Our thermodynamic characterization of site-directed mutated EF-hand motifs could contribute to better understand the structural factors that affect the cooperativity. EF-hand motifs are able to chelate a wide set of metal ions [38, 44] and our study also opens new perspectives for the optimization of efficient and specific metal ion binding sites by mutating EF-hand motifs of calmodulin.

Materials and methods

Protein engineering, expression and purification

The CaM-D1 construct containing the *Arabidopsis thaliana* sequence of the CaM N-terminal domain was

Table 1 Amino acid sequences for the wild-type CaM N-terminal domain (CaM wt I-II) and its mutants investigated in this work

Peptide		Site I													Site II												
		1	2	3	4	5	6	7	8	9	10	11	12		1	2	3	4	5	6	7	8	9	10	11	12	
CaM wt I-II	MADQLTDDQISEFKEAFSLF	D	K	D	G	D	D	G	C	I	T	T	K	E	LGTVMRSLGQNPTEAELQDMINEV	D	A	D	G	N	G	T	I	D	F	P	E	FLNLMARK
CaM Y I	MADQLTDDQISEFKEAFSLF	D	K	D	G	D	D	G	Y	I	T	T	K	E	LGTVMRSLGQNPTEAELQDMINEV	A	A	A	G	N	G	T	I	D	F	P	E	FLNLMARK
CaM Y II	MADQLTDDQISEFKEAFSLF	A	K	A	G	D	D	G	Y	I	T	T	K	E	LGTVMRSLGQNPTEAELQDMINEV	D	A	D	G	N	G	T	I	D	F	P	E	FLNLMARK
CaM Y I-II	MADQLTDDQISEFKEAFSLF	D	K	D	G	D	D	G	Y	I	T	T	K	E	LGTVMRSLGQNPTEAELQDMINEV	D	A	D	G	N	G	T	I	D	F	P	E	FLNLMARK
CaM YW II	MADQLTDDQISEFKEAFSLF	A	K	A	G	D	D	G	Y	I	T	T	K	E	LGTVMRSLGQNPTEAELQDMINEV	D	A	D	G	N	G	W	I	D	F	P	E	FLNLMARK
CaM YW I-II	MADQLTDDQISEFKEAFSLF	D	K	D	G	D	D	G	Y	I	T	T	K	E	LGTVMRSLGQNPTEAELQDMINEV	D	A	D	G	N	G	W	I	D	F	P	E	FLNLMARK

Residues that are modified (in comparison to the wild type sequence) are reported in red. Residues from the two binding sites are enclosed in the blue boxes. CaM Y I is impaired in binding Ca(II) at site II, CaM Y II and CaM YW II are impaired in binding Ca(II) at site I, whereas CaM Y I-II and CaM YW I-II are able to bind Ca(II) at both sites. CaM YW II and CaM YW I-II present a tryptophan residue at position 7 of site II Ca(II)-binding loop; all mutants present a tyrosine residue at position 7 of site I Ca(II)-binding loop

obtained as previously described [45], and used as a template for new constructs. In the original sequence, Le Clainche et al. [45] introduced a tyrosine at position 7 of site I binding loop in the peptides. In the present work, we also introduced a tryptophan at position 7 of the site II binding loop for the mutants CaM YW II and CaM YW I-II. In addition, we produced the Asp58Ala and Asp60Ala mutations in the CaM Y I construct to inactivate the metal-binding site II and the Asp22Ala and Asp24Ala mutations in CaM Y II and CaM YW II constructs to inactivate the metal-binding site I. All mutations were produced with the QuickChange site-directed mutagenesis kit (Stratagene) and specific primer pairs (see Supporting Information Table S4), according to the manufacturer's instructions. Subsequently, each mutated DNA sample was amplified with the D1-CaM-TEV-S and D1-CaM-STOP-AS primers to introduce the TEV protease recognition site upstream of the coding sequence as described in [46]. PCR products were amplified from 250 ng of linearized template DNA with 2.5 U of Taq polymerase (Prime Star; Lonza) for 30 cycles under the following cycling conditions: denaturation at 98 °C for 10 s, hybridization at 55 °C for 10 s and polymerization at 72 °C for 20 s. DNA fragments (282 bp) containing each coding sequence were isolated after digestion with *Bam*HI and *Pst*I, sequenced, and finally ligated into the *Bam*HI and *Pst*I sites of the pQE30 plasmid to obtain the PQE-plasmids for each mutant. Recombinant fusion proteins expressed in the *E. coli* strain M15Rep4 (Qiagen) were grown at 37 °C in LB medium containing ampicillin (50 mg/mL) and kanamycin (50 mg/mL). Expression was induced by adding 0.1 mM isopropyl-D-thiogalactoside once OD_{600nm} reached 0.5, and the cultures were further incubated for 5 h at 37 °C. Cellular extracts were obtained by French press lysis and a 30 min centrifugation step at 15,000 rpm. Extracts were then chelated with a 1 mL/min flow rate on a 5 mL HiTrap Chelating Column (GE Healthcare) in buffer A (50 mM Tris–HCl, 0.5 M NaCl, 25 mM imidazole, pH = 7.5) containing 1 mM AEBSF. The proteins were eluted from the nickel resin with a 4 mL/min flow rate using buffer A supplemented with 150 mM imidazole. Proteins were then dialyzed against buffer A, and the His-tag was removed by overnight incubation at 4 °C with TEV protease, followed by separation using a HiTrap Chelating Column. Recombinant proteins were dialyzed against 50 mM Tris–HCl, 150 mM NaCl (pH = 7.5). All protein samples were then electrophoresed on a 12 % Tris Tricine gel and stained with Coomassie Blue. Protein concentrations were measured according to the BC Assay (Uptima) with bovine serum albumin as the standard. The proteins were concentrated using the Microcon filtration system (Amicon Millipore®), with a cutoff point of 3 kDa.

Chemicals and stock solutions

MES buffer solution (20 mM, pH = 6) was prepared by weighting the appropriate amount of analytical grade salt (99.5 %, Sigma) and dissolving it in pure water. Solid KCl was added to the buffer solution to yield a concentration of 100 mM. A stock solution of calcium chloride was prepared by dissolving the appropriate amount of analytical grade $CaCl_2{\cdot}2H_2O$ (99.5 %, Merck) in pure water. All solutions were prepared with ultrapure water (18 MΩ). The pH values of the solutions were measured with a pH electrode (IoLine, Shott Instruments) and pH meter (Mettler Toledo); the electrode was calibrated with standard buffers. The pH values of buffer solutions were adjusted

with NaOH (10 M, Sigma) or HCl (37 %, Sigma). Before use, the buffer solutions were incubated for 1 h with Chelex-100 to remove any trace of calcium. For the same reason, after purification, the protein solutions were incubated for 3 h with a 100-fold excess of EDTA and subsequently dialysed against the calcium-free buffer solution (MES 20 mM, KCl 100 mM), to remove Ca(II)-EDTA complex as well as free EDTA. We used only plastic laboratory equipment to avoid any possible calcium contamination from glass. The residual concentration of Ca(II) in the buffer solutions was checked by ICP-AES. We did not detect any trace of the metal ion. The detection limit of the method was 25 nM, i.e., well below the Ca^{2+} concentration that may significantly affect CaM-Ca binding experiments.

Mass spectrometry analyses

Mass spectrometry (MS) analyses were performed on a MicroTOF-Q (Bruker) with an electrospray ionization source (ESI). CaM peptides were diluted in CH_3CN/H_2O (1/1 v/v), 0.2 % formic acid. Samples were continuously infused at a 3 µL/min flow rate. Mass spectra were recorded in the 50–7,000 mass-to-charge (*m*/*z*) range. MS experiments were carried out with a capillary voltage set at 4.5 kV and an end plate off-set voltage of 500 V. The gas nebulizer (N_2) pressure was set at 0.4 bar and the dry gas flow (N_2) at 4 L/min at 190 °C. Data were acquired in the positive mode and calibration was performed using a calibrating solution of ESI Tune Mix (Agilent) in CH_3CN/H_2O (95/5 v/v). The system was controlled by the software package MicroTOF Control 2.2 and data were processed with Data Analysis 3.4.

Circular dichroism measurements and thermal unfolding

The protein secondary structure and thermal unfolding were monitored by Far-UV circular dichroism spectroscopy, using a Jasco J-810 CD spectrometer equipped with a Jasco PTC-423 Peltier controller. CD spectra were recorded between 190 and 260 nm at a 1 nm resolution using 0.2 cm pathlength cuvettes. The CaM peptides were at 15.6 µM concentration in 20 mM Tris pH 7, and spectra were recorded without or with 440 µM $CaCl_2$ (Figure S7). Thermal denaturation curves were obtained by measuring the ellipticity at 222 nm from 23° to 95 °C in 0.2 °C increments, with a temperature slope of 1 °C per min and an averaging of 5 s. The CaM peptides were at 23 µM concentration in 10 mM Tris buffer pH 7, 100 mM NaCl (Figure S8). The thermal unfolding curves were fit to determine the melting temperature, Tm, as described in [47], by assuming a two-state equilibrium between native and unfolded forms of the protein.

Spectrofluorimetric titrations

The peptides' binding affinity for calcium was measured by monitoring the fluorescence intensity of a single aromatic residue; this was Tyr28 for CaM Y II and CaM Y I-II mutants, or Trp64 for CaM YW II and CaM YW I-II mutants. Spectrofluorimetric titrations were performed at pH = 6 and ionic strength $I = 0.12$ M (MES buffer, KCl). For each titration, a 10 µM peptide solution was prepared with MES/KCl buffer. Spectra were acquired both in the absence of and after stepwise additions of increasing $CaCl_2$ amounts. The fluorescence intensity of either the tyrosine residue (302 nm) or tryptophan residue (350 nm) was monitored during titrations. Spectra were collected on a Cary Eclipse spectrofluorimeter at 25 °C, with excitation at 270 nm for tyrosine and 260 nm for tryptophan. Emission was observed from 290 to 400 nm. The excitation and emission slits were 10 nm. The reported stability constants are averages of three experimental values.

ITC titrations

Isothermal titration calorimetry experiments were performed at 298 K using a MicroCal iTC200 device (GE Healthcare Life Sciences). The instrument consists of two identical cells, one for the sample and one for the reference solution. The reference cell of the microcalorimeter was filled with ultrapure water, and both cells were maintained at the same temperature. 200 µL of the protein solution (at concentrations ranging from 0.18 to 0.35 mM) was deposited in the sample cell during a typical titration, with stirring at 1000 rpm; 2 µL of the aqueous solution of $CaCl_2$ (with concentrations ranging from 2.5 to 5.0 mM) was injected at equal 150 s intervals (0.4 µL on the first injection). In total, 38.4 µL of $CaCl_2$ solution was added to the protein solution. When the $CaCl_2$ solution was injected into the protein solution, the heat of reaction was calculated at each injection as the difference in heat necessary to maintain the sample cell and reference cell at the same temperature. The heat effects due to dilution and mixing were assessed by injecting the $CaCl_2$ solution into the buffer solution under the same conditions, and these values were subtracted from the effects produced by the Ca(II)-protein reaction. The heat released by the dilution of the protein was very small, and was thus neglected. For consistency with the fluorescence experiments, ITC measurements were conducted at pH = 6 and ionic strength $I = 0.12$ M (MES buffer, KCl). Calorimetric analyses for the one-site peptides were evaluated with the "one site model" of the Origin™ fitting package, while the binding polynomial [48] was used for the two-sites peptides. The reported thermodynamic parameters are averages of three independent experimental values.

Results

We analyzed the affinity of the CaM N-terminal domain for Ca(II) ion. The interactions of Ca(II) with sites I and II and the cooperative effect between the binding sites were investigated using spectrofluorimetric and calorimetric measurements. Experimental results were corroborated by molecular dynamics analysis of the binding processes (see Supporting Information S3).

Five mutants of the recombinant N-domain of CaM (corresponding to a 77-amino-acid peptide) were investigated (Table 1): two "one-site" peptides (CaM Y II and CaM YW II), in which the metal-binding ability of site I was impaired by introducing the Asp22Ala and Asp24Ala point mutations, one "one-site" peptide (CaM Y I), in which binding at site II was impaired by introducing the Asp58Ala and Asp60Ala mutations [46], and two "two-sites" peptides (CaM Y I-II and CaM YW I-II) in which both sites I and II were able to chelate Ca(II). The fluorescence analyses on CaM Y I [46], CaM Y II and CaM YW II (see below) demonstrated that the suppression of two of the calcium ligands at positions 1 and 3 of the binding loop was sufficient to impair calcium binding at the mutated sites. These data are in line with literature data on troponin C, showing that mutation of one of the aspartate ligands was sufficient to impair calcium binding at site II [49].

The metal-binding isotherms were determined by following tyrosine or tryptophan luminescence at 302 or 350 nm, respectively. Tyr28 residue located at position 7 in the metal-binding loop of site I was sensitive to Ca(II) binding at site I and at site II since we observed a fluorescence emission change of Tyr28 upon metal ion binding at site II in the CaM Y II peptide. Fluorescence of Tyr28 could thus be used as a probe of metal binding at both sites I and II in the CaM Y I-II peptide. For the peptides CaM YW II and CaM YW I-II, a tryptophan was introduced at position 7 of the metal-binding loop of site II (mutation Thr-64Trp), in addition to the mutation Cys28Tyr. As such, it was possible to evaluate the metal-binding affinity of site II by following tryptophan fluorescence emission at 350 nm. As observed for Tyr28, Trp64 was also sensitive to metal binding at site I and could be used as a probe of metal binding at both sites I and II in the CaM YW I-II peptide. In addition, comparison of data obtained with CaM Y I-II and CaM YW I-II peptides allowed us to analyze the effect of the Thr64Trp mutation on microscopic binding constants and on cooperativity.

The recombinant peptides were produced in *E. coli* cells, as detailed in the Experimental section. A His-tag followed by the Tobacco Etch Virus protease (TEV) recognition sequence was introduced at the N-terminus, allowing the purification of the peptides using two subsequent chromatography steps on a Ni-column, as we previously described [46]. The purity of the produced peptides was verified by ESI–MS.

Spectrofluorimetric titrations

Spectrofluorimetric analyses were performed at pH = 6 and ionic strength $I = 0.12$ M. All investigated peptides display an increase in light emission upon addition of Ca(II), as previously described in the literature for CaM C-terminal domain [50]. The spectra collected during titrations are reported in Figure S5 of Supporting Information.

One-site peptides

Spectrofluorimetric titrations of CaM Y II and CaM YW II with Ca(II) provide monophasic binding isotherms (Fig. 2a, b), which represent the formation of 1:1 complexes (denoted as MP_T) from the reaction between the free calcium (M_f) and the free peptide (P_f) at site II, according to reaction (1)

$$M_f + P_f \overset{K_1}{\rightleftarrows} MP_T. \quad (1)$$

The binding isotherms have been fitted to a one site-binding model [Eq. (2)]

$$\Delta F/C_P = \Delta\phi K_1[M]/(1 + K_1[M]). \quad (2)$$

where C_P is the total protein concentration; $[M]$ is the free metal concentration, evaluated using an iteration procedure, as reported in Supporting Information S1; $\Delta F = (F - \phi_P C_P)$ is the change in fluorescence during titration, with F being the fluorescence signal; $\Delta\phi = (\phi_{MP} - \phi_P - \phi_M)$ is the amplitude of the binding isotherm, with ϕ_i being the molar fluorescence coefficient for the species i, and K_1 is the equilibrium constant of reaction (1). The monophasic binding isotherms show that only site II can bind calcium in these peptides. In this case, the measured macroscopic constant corresponds to the microscopic constant of the active binding site (Scheme 1):

$$K_1 = k_{II}. \quad (3)$$

The microscopic affinity constant was approximately twofold higher for the peptide containing the mutation Thr-64Trp, with $K_1 = 1.8 \times 10^4\ M^{-1}$ and $K_1 = 3.1 \times 10^4\ M^{-1}$ for CaM Y II and CaM YW II, respectively (corresponding to a dissociation constant K_d of 55 and 32 μM, respectively, Table 2). This demonstrates that the Thr64Trp point mutation increases the affinity of site II for Ca(II). The affinity constant value that we reported for site I in the one site peptide CaM Y I in a previous study [46] is about

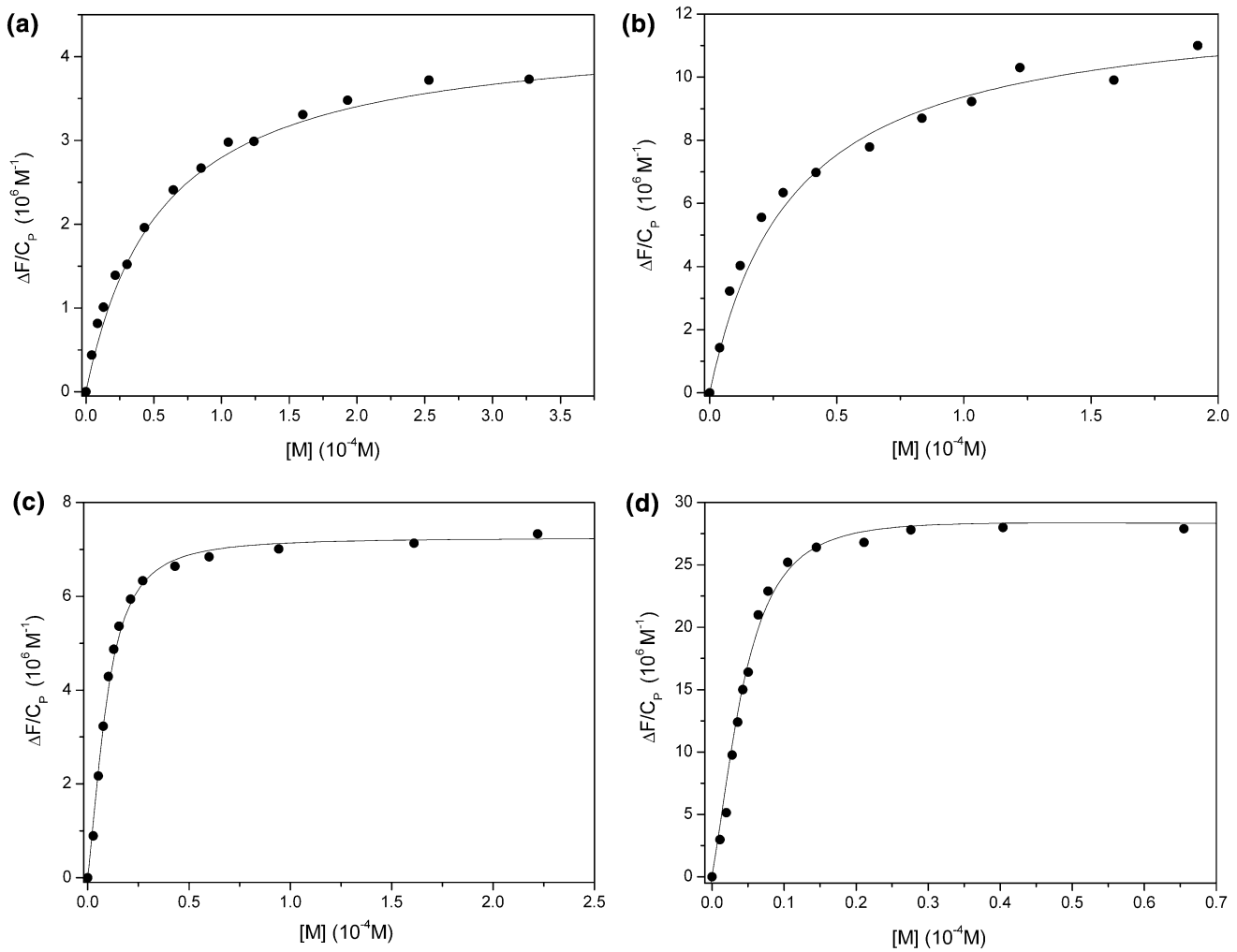


Fig. 2 Binding isotherms for the interaction of Ca(II) ion with **a** CaM Y II, **b** CaM YW II, **c** CaM Y I-II, **d** CaM YW I-II. $\lambda = 302$ nm for titrations (**a**) and (**c**), $\lambda = 350$ nm for titrations (**b**) and (**d**); $C_P = 1 \times 10^{-5}$ M; pH = 6 and $I = 0.12$ M (MES, KCl); $T = 298$ K. Data points are the average of three experiments. *Continuous lines* represent the best fits, according to Eq. (2) for panels (**a**) and (**c**) or Eq. (5) for panels (**b**) and (**d**)

1,5 times higher than that observed for site II in CaM Y II ($K_1 = 2.6 \times 10^4$ M^{-1}, corresponding to $K_d = 38$ μM).

Two-site peptides

Fluorescence analyses of Ca(II) binding to the two-site peptides CaM Y I-II and CaM YW I-II reveal binding isotherms that are well fitted by a two-site model (Fig. 2c, d). This confirms, for these peptides, that two binding steps occur, yielding a 2:1 metal:protein complex (M_2P_T). Therefore, the binding process is described by reactions (1) and (4).

$$M_f + MP_T \overset{K_2}{\rightleftarrows} M_2P_T. \tag{4}$$

The titration curves were analyzed according to the two-site binding model described by Eq. (5) [51]

$$\Delta F/C_P = (\Delta\phi_1 K_1[M] + \Delta\phi_2 K_1 K_2[M]^2)/ \left(1 + K_1[M] + K_1K_2[M]^2\right). \tag{5}$$

where $\Delta\phi_1 = (\phi_{MPT} - \phi_{Pf} - \phi_{Mf})$, $\Delta\phi_2 = (\phi_{M2PT} - \phi_{Pf} - \phi_{Mf})$ and K_2 is the equilibrium constant for reaction (4). Table 2 reports the macroscopic constants K_1 and K_2 for the investigated systems. Both K_1 and K_2 assume higher values when the point mutation Thr64Trp is inserted, showing that the Trp64 residue positively influences the calcium-binding process.

Very similar affinity constants were obtained from spectrofluorimetric titrations of CaM Y I-II with Ca(II)

Table 2 Macroscopic binding constants for the reaction of Ca(II) ion with the five mutants of the CaM N-terminal domain

Peptide	K_1 (10^4 M^{-1})	K_2 (10^4 M^{-1})
CaM Y II	1.8 ± 0.3[a]	–
	1.9 ± 0.2[b]	
CaM YW II	3.1 ± 0.3[a]	–
	3.6 ± 0.5[b]	
CaM Y I	3.1 ± 0.4[b]	–
	2.6 ± 0.1[c]	
CaM Y I-II	5.3 ± 0.3[a]	14 ± 3.0[a]
	6.0 ± 1.0[b]	8.0 ± 2.0[b]
CaM YW I-II	8.1 ± 0.4[a]	31 ± 3.0[a]
	6.1 ± 1.1[b]	15 ± 5.0[b]

pH = 6 and $I = 0.12$ M (MES, KCl); $T = 298$ K. All reported data are the average of three experimental values

[a] Data from spectrofluorimetric titrations

[b] Data from calorimetric titrations

[c] Binding constant obtained from spectrofluorimetric titration in a previous study [46]

performed at pH = 7.5 ($K_1 = 5.8 \times 10^4 \pm 0.4 \times 10^4$ M^{-1}; $K_2 = 1.0 \times 10^5 \pm 0.3 \times 10^5$ M^{-1}, Figure S6) and at pH = 6 (Table 2). Such a result is in accordance with the fact that no residue of CaM site I or site II is subjected to acid dissociation in the pH range between 6 and 7.5.

The macroscopic constant for the second metal ion binding, K_2, is significantly higher than K_1, which strongly suggests cooperativity, as reported for Ca(II) binding to bovine calmodulin [20]. We used the Scatchard plot [52] and the Hill plot [53] to analyze spectrofluorimetric data, to evaluate the cooperative effect upon Ca(II) binding to CaM Y I-II and CaM YW I-II (see Supporting Information S2). The obtained concave downward Scatchard plots, with an intercept value $B_{max} = 2$ (Figure S2.1), and the Hill coefficients $n_H > 1$ (Figure S2.2) are representative of positive cooperativity between protein-binding sites I and II for the peptides CaM Y I-II and CaM YW I-II. Nevertheless, these approaches do not give quantitative information regarding how ligand binding at one site influences the affinity of the other site. Therefore, we used a more accurate data treatment to analyze cooperativity, based on the evaluation of $\Delta\Delta G$. This parameter represents the difference in free energy between calcium binding to apo-CaM and calcium binding to CaM loaded with one Ca(II) ion [20]. In addition to describing the type of cooperativity (with $\Delta\Delta G < 0$ corresponding to a positive cooperativity and $\Delta\Delta G > 0$ corresponding to a negative one), $\Delta\Delta G$ also allows the quantitative measurement of the cooperative effect.

To evaluate the free energy of site–site interaction, $\Delta\Delta G$, the microscopic binding constants, defined in Scheme 1, are necessary, as $\Delta\Delta G$ is defined as:

$$\Delta\Delta G = -RT \ln\left(k_{I,II}/k_I\right) = -RT \ln\left(k_{II,I}/k_{II}\right). \quad (6)$$

On the other hand, only macroscopic binding constants are evaluated from experimental analysis of the N-terminal domain. To the best of our knowledge, only the lower limit of the free energy of interaction ($\Delta\Delta G = -RTln(4K_2/K_1)$) has been calculated for the CaM-Ca(II) interaction using such macroscopic binding constants, by assuming that site I and site II have an identical affinity for the ligand [20, 29, 32, 35]. The hypothesis underpinning this estimate is not necessarily satisfied in real systems, as different sites of a protein often do not show the same exact affinity for the same ligand.

To obtain microscopic binding constants, we analyzed one-site peptides obtained by site-directed mutagenesis of calcium-binding ligands located at positions 1 and 3 of the Ca(II) binding loops of site I (CaM Y II and CaM YW II) or site II (CaM Y I). The substituted alanine residues impair metal binding [46, 49]. In addition, these residues located at the N-terminal part of the loop are not involved in the site–site interaction, unlike other residues such as amino acids 7–9 of the loops, involved in an antiparallel sheet structure that allows cooperativity of Ca(II) binding [54]. The ligands of the N-terminal part of the loop were reported as highly mobile and to fold around the metal ion without significant effect on the domain structure [23]. Accordingly, we observed that the Far-UV CD spectra of the different peptides CaM Y I, CaM Y II and CaM Y I-II were superimposable, in the absence and in the presence of Ca(II) (Supplementary Fig. S7). Comparable structural models were also obtained by molecular dynamic simulations for the Ca(II)-free site I of CaM Y I and CaM Y II and that of wild-type CaM (not shown). Ca(II)-binding affinities in EF-hand peptides also depend on the stability of the protein. To evaluate a possible impact of site-directed mutagenesis on the stability of the apo-proteins, we compared the thermal unfolding properties of peptides CaM Y I, CaM Y II and CaM Y I-II using circular dichroism measurements (Supplementary Fig. S8). Similar denaturation curves were obtained with the three CaM peptides and the melting temperatures differed by less than 2 °C between CaM Y I (Tm = 58 °C) and CaM Y I-II (Tm = 59.8 °C) and by 3 °C between CaM Y II (Tm = 62.9 °C) and CaM Y I-II. Consequently, we made the assumption that the observed binding process at site II for the one site peptides CaM Y II and CaM YW II is similar to the binding process at site II of the corresponding two-site mutants. Following this hypothesis, we replaced k_{II} of CaM Y I-II and CaM YW I-II with the experimental values obtained for CaM Y II and CaM YW II, respectively. These k_{II} values were introduced in equations reported in Scheme 1, together with the K_1 and K_2 values obtained from the analysis of experimental data according to Eq. (5). Values of $\Delta\Delta G$ and of the microscopic constants k_I, $k_{I,II}$ and $k_{II,I}$ are reported in Table 3.

It is important to note that the microscopic constant k_I, when calculated this way, is very close to the constant

Table 3 Cooperativity parameters and microscopic binding constants for the reaction of Ca(II) ion with the two-site mutants of the CaM N-terminal domain, evaluated using spectrofluorimetric experimental data

Peptide	n_{Hill}	$\Delta\Delta G$ (kJ mol^{-1})	k_I (10^4 M^{-1})	k_{II} (10^4 M^{-1})	$k_{II,I}$ (10^4 M^{-1})	$k_{I,II}$ (10^4 M^{-1})
CaM Y I-II	1.8 ± 0.03	−6.1 ± 0.1	3.5 ± 0.1	1.8 ± 0.3	21 ± 2	41 ± 3
CaM YW I-II	1.9 ± 0.02	−6.9 ± 0.1	5.0 ± 0.1	3.1 ± 0.3	50 ± 3	81 ± 3

pH = 6 and I = 0.12 M (MES, KCl); T = 298 K

measured experimentally for the one-site peptide CaM Y I (Table 2), in which site II was impaired by Asp58Ala and Asp60Ala mutations [46]. This supports the view that mutations at positions 1 and 3 of the metal-binding loop in either site I or site II do not significantly alter the properties of calcium binding for the other binding site, and that the microscopic values obtained using one-site peptides allow an accurate estimation of inter site cooperativity.

The $\Delta\Delta G$ values of Table 3 show a positive inter site cooperativity for calcium binding at the N-terminal domain of calmodulin, which is reinforced by the presence of a tryptophan at position 7 of the binding loop of site II.

ITC titrations

ITC measurements of Ca(II) binding to all analyzed peptides were made to evaluate the enthalpy and entropy changes related to these reactions, as well as the binding constants. Isothermal calorimetric titrations were performed at pH = 6 and ionic strength I = 0.12 M (MES, KCl), by adding increasing amounts of $CaCl_2$ to a solution of the peptide.

One-site peptides

For the one-site peptides CaM Y I, CaM Y II and CaM YW II, the binding isotherms were best described by a "one set of sites" binding model. The equilibrium constants obtained for Ca(II) binding at site I (CaM Y I), $K_1 = 3.1 \times 10^4$ M^{-1}, and at site II for CaM Y II and CaM YW II, $K_1 = 1.9 \times 10^4$ M^{-1} and $K_1 = 3.6 \times 10^4$ M^{-1}, respectively, were in good accordance with those obtained by spectrofluorimetric analysis (Table 2). These equilibrium constants and the protein concentration used in the experiments explain why the binding isotherms reached their endpoints at a Ca(II) to protein molar ratio of about 2.5. The enthalpy change associated with Ca(II) binding at site I was negative, indicative of an exothermic process (Fig. 3). In contrast, the enthalpy change associated with Ca(II) binding at site II in CaM Y II and CaM YW II was positive, indicative of an endothermic process (Fig. 4a, b). This enthalpy change ΔH_1 was three times larger for Ca(II) binding at site II in the Thr64Trp mutant (CaM YW II peptide).

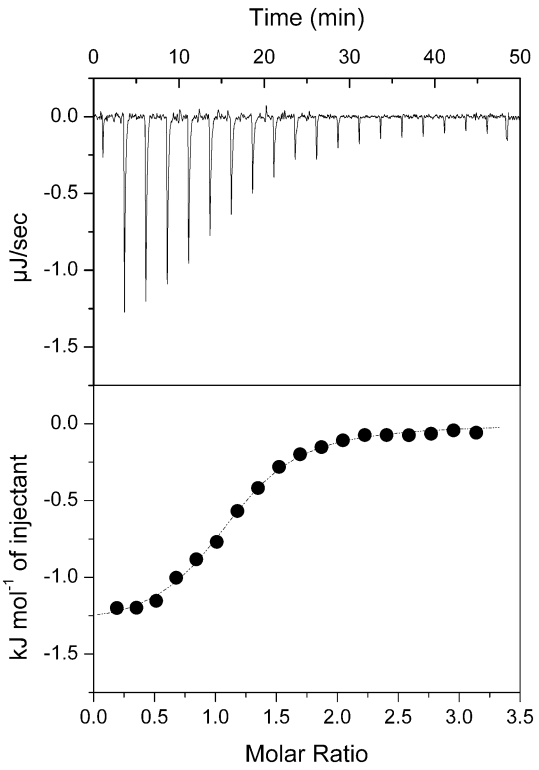


Fig. 3 ITC raw data and binding isotherm for Ca(II) ion interaction with 0.35 mM CaM Y I; pH = 6 and I = 0.12 M (MES, KCl); T = 298 K. Data points are the average of three experiments. *Dashed line* represents the best fit, according to a "one set of sites" model

However, with ΔH_1 values of ~−1.3 kJ mol^{-1} upon Ca(II) binding to site I in the CaM Y I peptide or ΔH_1 values of ~1 and 3 kJ mol^{-1} for Ca(II) binding at site II in CaM Y II and CaM YW II respectively, the enthalpy contribution to the free energy of binding is small, and ten to twenty times less than the entropic term $T\Delta S_1$, calculated from measured values of K_1 and ΔH_1 (Tables 2, 4). Binding of calcium is thus mainly driven by entropy.

Two-site peptides

For the two-site peptides, the biphasic nature of the isotherms clearly pointed to the existence of two non-equivalent binding steps (Fig. 4c, d). The general binding polynomial for a two-site system, reported in Eq. (7), was used

to evaluate the heat associated with each injection and to fit the experimental curves through a nonlinear least squares regression analysis as described in [55].

$$X = 1 + K_1[M] + K_1K_2[M]^2. \quad (7)$$

All thermodynamic parameters obtained from the calorimetric analysis of these reactions are reported in Tables 2 and 4.

An endothermic process prevails at low Ca(II) concentrations, while at higher Ca(II) concentrations, the binding curve becomes clearly exothermic (Fig. 4c, d). The experimental data can be fit with macroscopic binding constants of about $6.0 \pm 1.0 \times 10^4$ M^{-1} for the first binding event for both CaM Y I-II and CaM YW I-II and of $8.0 \pm 2.0 \times 10^4$ and $15 \pm 5.0 \times 10^4$ M^{-1} for CaM Y I-II and CaM YW I-II, respectively, for the second exothermic binding event (Table 2). These macroscopic binding constants slightly differ from those obtained by fluorescence spectroscopy. These differences may be due to differences in experimental conditions and a larger imprecision of the fit for the ITC data, possibly explained by the low enthalpy values obtained upon Ca(II) binding. Indeed, as already noticed for Ca(II) binding in one-site peptides, the absolute values of ΔH are small and the enthalpy contribution to the Gibbs free energy of binding very small in comparison with the entropic term, with ΔS values ranging from 90.6 to 103.8 J mol^{-1} K^{-1} (Table 4). The thermodynamic parameters obtained in this study are of the same order of magnitude than those reported for the N-terminal domain of human calmodulin that was fitted with a one-site model ($\Delta H = 7.022 \pm 0.25$ kJ Mol^{-1} and $\Delta S = 124{,}14$ J mol^{-1} K^{-1}, [38]).

For both two-site mutants, there is a significant difference between the ΔH values of the first and second binding steps, with $\Delta H_1 = 1.5$ kJ mol^{-1} and $\Delta H_2 = -1.0$ kJ mol^{-1} for CaM Y I-II and $\Delta H_1 = 3.7$ kJ mol^{-1} and $\Delta H_2 = -4.6$ kJ mol^{-1} for CaM YW I-II (Table 4). The presence of a tryptophan residue at position 7 of the Ca(II) binding loop of site II noticeably enhanced the enthalpy gain of the interaction, for both the first and second binding steps (Table 4).

Discussion

Evaluation of the site-specific binding constants

The goals of this study were to evaluate the specific calcium affinity of the structured EF-hand motifs of site I and site II and to obtain an accurate thermodynamic description of the cooperative Ca(II) binding of the isolated CaM N-terminal domain. The estimation of microscopic binding constants of the individual EF-hand motifs is complicated by the contribution from the cooperative effect between the binding sites. As recently pointed out by Haiech et al. [9], deconvolution of the contribution of each site to the CaM-Ca(II) binding process is still challenging. Numerous studies devoted to Ca(II) interactions with calmodulin [20, 32, 33, 56, 57] or other EF-hand proteins [58, 59] provide information about macroscopic binding affinity, but they do not give an evaluation of site-specific binding constants. More recently, Ye et al. [34] have made a step forward in this direction by grafting CaM single-binding loops in a scaffold protein. This method gave useful information on the intrinsic calcium-binding properties of the binding loops, because the grafting approach preserved the structure of these loops, and thus avoided the problem of low cation-binding affinities of model peptides associated with the lack of pre-defined structure in solution. In this respect, work performed on cyclic peptides derived from calmodulin presented also interesting information on the role of each calcium-binding ligand on calcium or metal ions affinity in calmodulin [60].

In this study, we wanted to take into account the role of flanking regions of the cation-binding loop in defining the specific metal site properties. The site-directed mutagenesis approach allowed us to take into account longer distance interactions with the metal-binding site and thermodynamic effects of the whole EF-hand motif in the allosteric process. Working on the isolated N-terminal domain, and not on the entire calmodulin, allowed us to circumvent inter domain effects, which largely impact the Ca^{2+} affinity for the N-terminal domain due to preferential binding of Ca^{2+} to the C-terminal domain, and formation of a high-affinity conformation [61]. Working on the isolated N-terminal domain explains why the dissociation constants are an order of magnitude lower than those assigned to site I and site II in the whole protein [61, 62]. Using site-directed mutagenesis, and making the assumption that the binding properties of Ca(II) to site I or site II do not differ significantly in the one-site and in the two-site peptides, we obtained good estimates of the partially Ca(II)-occupied states of the CaM N-terminal domain, of the intra domain cooperativity, and information on the structure–affinity relationship of each binding site. This method has also been used to study metal binding to calbindin, another EF-hand protein, although not all the microscopic constants in the system were evaluated [58, 59].

For the two-site peptides, we determined that site I binds Ca(II) slightly more efficiently than site II, with k_I almost twofold higher than k_{II} for CaM Y I-II ($k_I = 3.5 \times 10^4$ M^{-1} and $k_{II} = 1.8 \times 10^4$ M^{-1}). The same trend in affinity constant values ($k_I > k_{II}$) was reported by Ye et al. [34] for the grafted single sites I and II of CaM, as well as for sites I and site II in the high-affinity conformation of the whole calmodulin [61]. Nevertheless, the affinities of the grafted

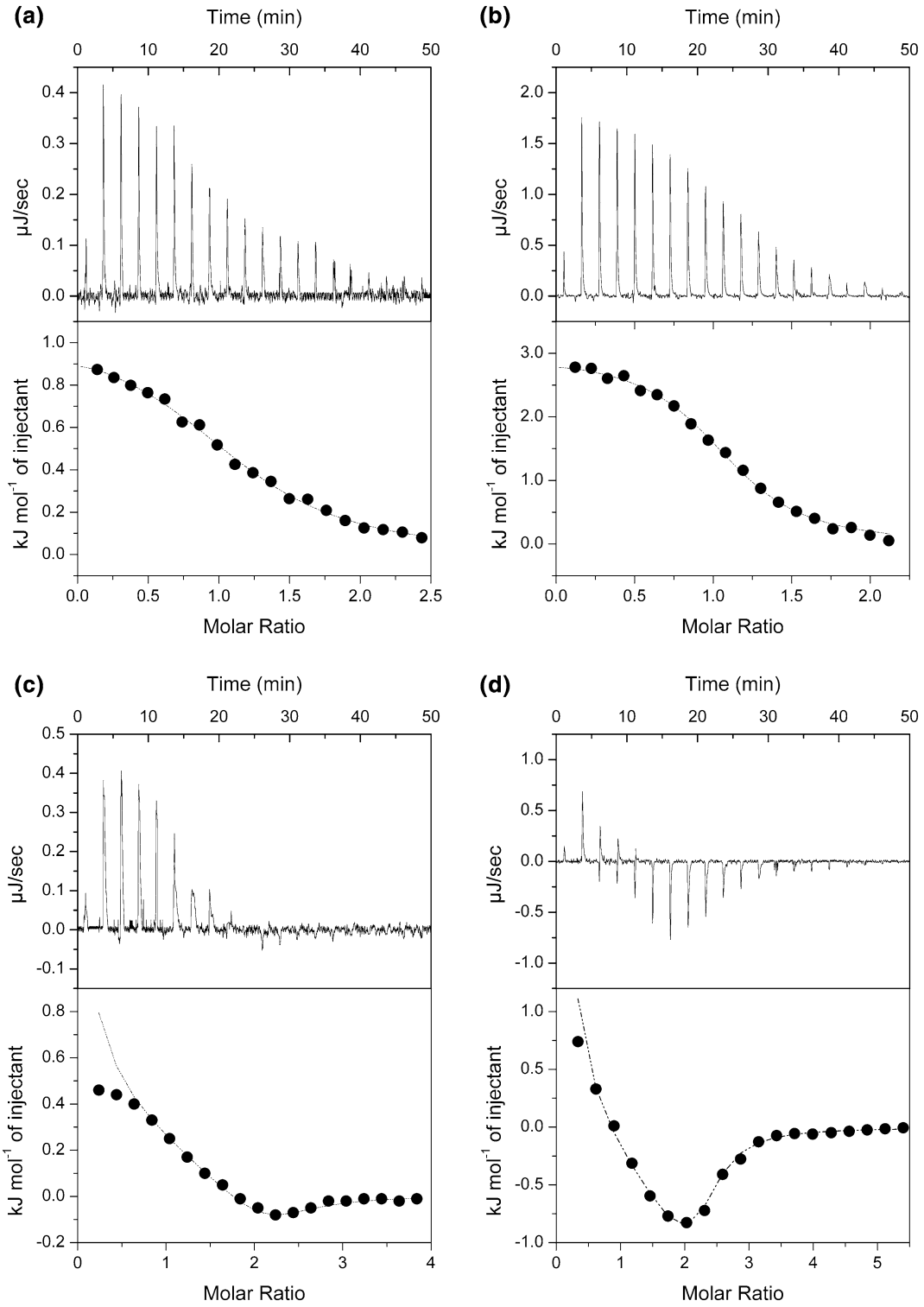


Fig. 4 ITC raw data and binding isotherms for reactions: **a** 0.20 mM CaM Y II with 2.5 mM $CaCl_2$, **b** 0.28 mM CaM YW II with 3.0 mM $CaCl_2$, **c** 0.20 mM CaM Y I-II with 4.0 mM $CaCl_2$, **d** 0.16 mM CaM YW I-II with 4.5 mM $CaCl_2$; pH = 6 and $I = 0.12$ M (MES, KCl); $T = 298$ K. Data points are the average of three experiments. *Dashed lines* represent the best fits, according to a "one set of sites" model for panels (**a**) and (**b**) or a "two sites" binding polynomial for panels (**c**) and (**d**)

Table 4 Thermodynamic parameters of Ca(II) binding to the investigated mutants of the CaM N-terminal domain, obtained from ITC analysis

Peptide	ΔH_1 (kJ mol^{-1})	ΔS_1 (J mol^{-1}K^{-1})	$T\Delta S_1$ (kJ mol^{-1})	ΔG_1 (kJ mol^{-1})	ΔH_2 (kJ mol^{-1})	ΔS_2 (J mol^{-1}K^{-1})	$T\Delta S_2$ (kJ mol^{-1})	ΔG_2 (kJ mol^{-1})
CaM Y I	-1.3 ± 0.1	82.3 ± 0.9	24.5 ± 0.3	-25.8 ± 0.4	–			
CaM Y II	1.0 ± 0.1	85.4 ± 0.9	25.4 ± 0.3	-24.4 ± 0.4	–			
CaM YW II	3.0 ± 0.1	97.1 ± 1.1	28.9 ± 0.3	-25.9 ± 0.4	–			
CaM Y I-II	1.5 ± 0.2	96.6 ± 0.3	28.8 ± 0.3	-27.2 ± 0.2	-1.0 ± 0.2	90.6 ± 0.3	27.0 ± 0.3	-27.2 ± 0.2
CaMYWI-II	3.7 ± 0.5	103.8 ± 0.5	30.9 ± 0.5	-27.3 ± 0.2	-4.6 ± 0.5	96.1 ± 0.6	28.6 ± 0.6	-29.5 ± 0.3

pH = 6 and $I = 0.12$ M (MES, KCl); $T = 298$ K. The entropies of association were calculated from measured values of K and ΔH, using the equations: (1) $\Delta G = \Delta H - T\Delta S$ and (2) $\Delta G = -RTlnK$

loops ($k_I = 8.0 \times 10^3$, $k_{II} = 2.5 \times 10^3$ M^{-1}, 2 mM Tris, pH = 7.5, [KCl] = 150 mM) were four to seven fold weaker than the affinities presented here. This demonstrates that structural or longer range interactions within the EF-hand motifs significantly affect the Ca(II)-binding affinities.

The relative affinities of site I and II for Ca(II) obtained in this work are in good accordance with previous work, in which the affinity for Ca(II) was proposed to increase with the number of negatively charged carboxylic groups involved in the metal ion coordination [63]. A comparison of the primary sequences of the two binding sites allows to qualitatively assess that site I, with three aspartates and one glutamate ligands of Ca(II), is expected to have a higher affinity than site II, where the aspartate at position 5 of the loop is replaced by an asparagine residue. However, other factors as hydrogen bonding networks and electrostatic interactions are also important factors determining the intrinsic affinity of Ca(II)-binding sites.

This is illustrated by the fact that the difference in binding affinity of Ca(II) at site I and site II is smaller for the variant containing a tryptophan at position 7 of the binding loop of site II ($k_I = 5.0 \times 10^4$ and $k_{II} = 3.1 \times 10^4$ M^{-1}, respectively).

Cooperativity and conformational changes

Our results provide insight into the cooperativity of Ca(II) binding by the CaM N-terminal domain at a microscopic level. Evaluation of the microscopic constants for Ca(II) binding at each site of the peptides allowed us to calculate ΔΔG values and evaluate cooperativity in a more accurate way, as compared to the commonly used lower limit of ΔΔG. In the particular case of CaM N-terminal domain, we observed that the intrinsic Ca(II)-binding affinities for the two sites did not differ largely, and the ΔΔG value obtained in this study for CaM Y I-II ($-6.1 \pm 0{,}1$ kJ mol^{-1}) was on the same order of magnitude as the $\Delta\Delta G_{lim}$ estimated in previous studies of CaM N-terminal domains, where $\Delta\Delta G_{lim}$ ranged from −11.9 to −3.9 kJ mol^{-1}, depending on the fragments [29, 33, 35]. Conversely, the cooperative effect estimated by considering the Ca(II) affinities of CaM grafted sites (−28.6 kJ mol^{-1}, [34] is more than fourfold higher than what we estimated for the entire N-terminal domain (−6.1 kJ mol^{-1}). For the two-site peptides, we obtained a more than ten-fold higher affinity for the second metal ion that binds an already partially occupied peptide, with $k_{I,II}/k_I = k_{II,I}/k_{II} = 11.7$ for CaM Y I-II and $k_{I,II}/k_I = k_{II,I}/k_{II} = 16.1$ for CaM YW I-II. These results highlight a pronounced positive cooperative effect that is further significantly enhanced in the presence of the Thr64Trp mutation.

Calorimetric data provide interesting information regarding the interaction energy–structure relationship. The calorimetric studies of the CaM-Ca(II) interaction reported in literature [61, 64, 65] are controversial, as they provide very different values of ΔH and ΔS. Moreover, these analyses are related to the entire calmodulin, so it is not possible to attribute each effect to the individual binding sites. On the other hand, the analysis of one-site peptides allowed us to evaluate the thermodynamic parameters of Ca(II) binding to site I or to site II alone.

We observed that Ca(II) binding at site I is exothermic while Ca(II) binding at site II is endothermic. In fact, the ΔH values were very small, with values of −1.3 and 1.0 kJ mol^{-1}, for site I and site II respectively, and the ITC experiments showed that the enthalpic contribution to the free energy of binding is negligible in comparison to the entropic contribution.

This behavior is common for the binding of divalent cations to proteins due to their high dehydration free energy [41, 66]. For comparison, the Gibbs free energy of calcium hydration was reported at 597 kJ mol^{-1} at 298.15 K [67]. In addition to the gain in solvent entropy, metal binding at sites I and II may be driven by the reorganization of amino acid residues around the metal ion, which implies a high and favorable entropy change, especially at neutral pH, where a strong repulsion exists between the anionic charges of the ligands. Differences in the number and position of positively and negatively charged residues in the

metal binding loops of site I and site II may explain the slight differences in entropy observed upon Ca(II) binding ($\Delta S = 82.3$ J mol^{-1} K^{-1} and $\Delta S = 85.4$ J mol^{-1} K^{-1} for CaM Y I and CaM Y II, respectively). For site I, two positively charged lysines (Lys23 and Lys32) are present in addition to the four negatively charged aspartate and glutamate residues (Asp22, Asp24, Asp26, Glu33). Therefore weaker electrostatic repulsion may exist between the ligands, with respect to site II where only negatively charged ligands are present (Table 1). As discussed in Gifford et al. [68] however, many structural parameters may impact the entropic term, as hydrogen bonding interaction or the orientation of alpha helices, as well as the dynamics of the protein backbone.

The ITC analysis of the two-site mutants confirms that the calcium-binding events are mainly entropically driven. It also highlights the fact that the enthalpy term is more favorable for the second calcium-binding event than for the first Ca(II) binding, with $\Delta H_1 > 0$ and $\Delta H_2 < 0$ for both the two-site mutants (Table 4). Former NMR data on calbindin showed that the flexibility of the peptide significantly decreased upon binding of the first metal ion, and that both binding sites are already trapped in the final two-metal occupied conformation [69]. This open structure seems to confer a favorable binding enthalpy for the second Ca(II)-binding event.

Influence of the Thr64Trp point mutation on Ca(II) affinity and cooperativity

Bidentate binding of glutamate to calcium in the first site was pointed as the mechanistic event driving the change in orientation of the helices of both site I and site II towards the optimized conformation for the second metal binding [37–39]. Another factor that influences cooperativity is the inter-site β-sheet formed by residues 7 and 8 of the two calcium-binding loops [41, 42, 54]. In this study, we analyzed how the Thr64Trp mutation at position seven of site II binding loop influences site I and site II properties, as well as cooperativity.

The main influence of the Thr64Trp mutation is observed on Ca(II) binding at site II, where the k_{II} for CaM YW I-II was almost two-fold higher than the k_{II} for CaM Y I-II. The main chain carbonyl group of Trp64 is most probably involved in Ca(II) coordination in site II, as observed in the structural models obtained by a molecular dynamics approach (Fig. 1b, supporting information). The tryptophan side-chain probably modifies the strength of the Ca(II) carbonyl interaction. In addition, this difference in affinity may be due to the higher hydrophobicity of tryptophan side chain, when compared to threonine. Indeed, it has been reported for a variety of proteins that hydrophobic interactions favor the folding of the binding loop, which diminishes the distances between the ligands and the metal ion and consequently increases the affinity of the binding site [70, 71]. This hypothesis is in line with the observation that ethylene glycol largely increases the affinity of tryptophan-containing CaMs for Ca(II), because of a decrease in the dielectric constant of the solution [62]. A non-covalent cation-π interaction between Ca(II) and the indole ring of tryptophan could also account for the increase in affinity. Such an interaction can occur for cations that are within 6 Å of the face of an aromatic ring [31, 72]. We obtained an average Ca(II)-indole distance of 5.9 Å from our molecular dynamics simulations (see Supporting Information S3). This distance is considered as a theoretical upper limit value that does not take into account the hydrophobic effect, since the simulation was performed *in vacuo*. Hence, we can assume an even lower Ca(II)-indole distance in water solution.

In addition, it is worth noting that the Thr64Trp mutation significantly increases the Ca(II) affinity of site I as well, with $k_I = 3.5 \times 10^4$ M^{-1} and $k_I = 5.0 \times 10^4$ M^{-1} for CaM Y I-II and CaM YW I-II, respectively. This indirect effect is probably due to the fact that Trp64 (site II) and Tyr28 (site I) are part of an inter site β-sheet structure (see Fig. 1). Actually, the location of Trp64 on the inter site β-sheet probably explains not only its positive role on Ca(II) binding at site I, but also its positive role on the cooperativity between sites I and II, by tuning the hydrogen bonding properties of the backbone carbonyl groups of the amino acids involved in the β-sheet structure [54] and hence the interaction between the main chain carbonyl of the residue at position 7 of the loop and Ca(II), and the structural reorganization upon the first Ca(II) binding event.

Conclusion

A detailed thermodynamic reinvestigation of calcium binding to the N-terminal domain of calmodulin was performed in this work. Our results provided site-specific Ca(II)-binding constants and a quantitative measurement of the cooperative effect between the two sites of the isolated N-terminal domain. This approach also enabled an estimation of the influence of a point mutation—Thr64Trp-, on both the specific Ca(II)-binding affinity in site II, and the cooperativity. ITC measurements allowed us to resolve the enthalpic and entropic components of the binding affinity, which are intimately related to structural parameters.

By coupling site-directed mutagenesis with the thermodynamic analysis we could evaluate the microscopic constants for calcium binding to a free (non-grafted) CaM N-terminal domain. The advantage of such methodology lies in the possibility to take into account the influence of flanking regions and conformational changes associated

with metal binding. Another advantage of using non-grafted peptides consists in the possibility to analyze the partially occupied state and, therefore, the cooperative effect. Moreover, by mutating the residues of the N-terminal part of the loops, instead of those of the C-terminal part, we did not drastically change the structural and dynamical properties of the mutant peptides, compared to the wild-type sequence.

The findings of the present work represent a significant advance in rationalizing, at a microscopic level, the interactions of EF-hand proteins with metal ions, on which their biological role depends. However, it is worth to notice that our results concern the CaM N-terminal domain; therefore, some differences in the thermodynamic parameters and in the binding mechanism can be expected for the full-length protein.

Acknowledgments This work was funded in part by the Toxicologie program CalUMo from the CEA, by the ECCOREV Research federation, and by the Programme Investissement Avenir DEMETERRES. We acknowledge Pascale Delangle for useful discussions.

References

1. Popovych N, Shangjin S, Ebright RH, Kalodimos GC (2006) Nat Struct Mol Biol 13:831–838
2. Kern D, Zuiderweg ERP (2003) Curr Opin Struct Biol 13:748–757
3. Kuriyan J, Eisenberg D (2007) Nature 450:983–990
4. Cheung WY (1980) Science 207:19–27
5. Klee CB, Crouch TH, Richman P (1980) Annu Rev Biochem 49:489–515
6. Wang JH, Waisman DM (1979) Current Top Cell Regul 15:47–107
7. Torok K, Whitaker M (1994) BioEssays 16:221–224
8. Haiech J, Audran E, Fève M, Ranjeva R, Kilhoffer MC (2011) Biochimie 93:2029–2037
9. Haiech J, Gendrault Y, Kilhoffer MC, Ranjeva R, Madec M, Lallement C (2014) Biochim Biophys Acta 1843:2348–2355
10. Li W, Wang W, Takada S (2014) Proc Natl Acad Sci USA 111:10550–10555
11. Crivici A, Ikura M (1995) Annu Rev Bioph Biom 24:85–116
12. Rhoads AR, Friedberg F (1997) FASEB J 11:331–340
13. Jurado LA, Chockalingam PS, Jarrett HW (1999) Physiol Rev 79:661–682
14. Berchtold MW, Villalobo A (2014) Biochim Biophys Acta 1843:398–435
15. Babu YS, Sack JS, Greenhough TJ, Bugg CE, Means AR, Cook WJ (1985) Nature 315:37–40
16. Babu YS, Bugg CE, Cook WJ (1988) J Mol Biol 204:191–204
17. Kretsinger RH (1976) Annu Rev Biochem 45:239–265
18. McPhalen CA, Strynadka NCJ, James MNG (1991) Adv Protein Chem 42:77–144
19. Strynadka NCJ, James MNG (1989) Annu Rev Biochem 58:951–999
20. Linse S, Helmersson A, Forse S (1991) J Biol Chem 266:8050–8054
21. Kuboniwa H, Tjandra N, Grzesiek S, Ren H, Klee CB, Bax A (1995) Nat Struct Mol Biol 2:768–776
22. Zhangh M, Tanaka T, Ikura M (1995) Nat Struct Mol Biol 2:758–767
23. Grabarek Z (2005) J Mol Biol 346:1351–1366
24. Chou JJ, Li S, Klee CB, Bax A (2001) Nat Struct Biol 8:990–997
25. Evenas J, Malmendal A, Akke M (2001) Structure 9:185–195
26. Barbato B, Ikura M, Kay LK, Pastor RW, Bax A (1992) Biochemistry 31:5269–5278
27. Bayley PM, Martin SR (1992) Biochim Biophys Acta 1160:16–21
28. Klevit RE (1983) Methods Enzymol 102:82–104
29. Sorensen BR, Shea MA (1998) Biochemistry 37:4244–4253
30. Masino L, Martin SR, Bayley PM (2000) Protein Sci 9:1519–1529
31. Tsalkova TN, Privalov PL (1985) J Mol Biol 181:533–544
32. VanScyoc WS, Shea MA (2001) Protein Sci 10:1758–1768
33. Sorensen BR, Faga LA, Hultman R, Shea MA (2002) Biochemistry 41:15–20
34. Ye Y, Lee HW, Yang W, Shealy S, Yang JJ (2005) J Am Chem Soc 127:3743–3750
35. Waltersson Y, Linse S, Brodin P, Grundstrom T (1993) Biochemistry 32:7866–7871
36. Evenas J, Malmendal A, Thulin E, Carlstrom G, Forsen S (1998) Biochemistry 37:13744–13754
37. Gagne SM, Li MX, Sykes BD (1997) Biochemistry 36:4386–4392
38. Senguen FT, Grabarek Z (2012) Biochemistry 51:6182–6194
39. Maune JF, Beckingham K, Martin SR, Bayley PM (1992) Biochemistry 31:7779–7786
40. Browne JP, Strom M, Martin SR, Bayley PM (1997) Biochemistry 36:9550–9561
41. Malmendal A, Evenas J, Forsén S, Akke M (1999) J Mol Biol 293:883–899
42. Marchand S, Roux B (1998) Proteins 33:265–284
43. Grabarek Z (2006) J Mol Biol 359:509–525
44. Ouyang H, Vogel HJ (1998) Biometals 11:213–222
45. LeClainche L, Planque G, Amekraz B, Moulin C, Pradines-Lecomte C, Peltier G, Vita C (2003) J Biol Inorg Chem 8:334–340
46. Pardoux R, Sauge-Merle S, LeMaire D, Delangle P, Guilloreau L, Adriano JM, Berthomieu C (2012) PLoS One 7:e41922
47. Greenfield NJ (2006) Nat Protoc 1:2527–2535
48. Freire E, Schon A, Velazquez-Campoy A (2009) Method Enzymol 455:127–155
49. Putkey JA, Sweeney HL, Campbell ST (1989) J Biol Chem 264:12370–12378
50. VanScyoc WS, Sorensen BR, Rusinova E, Laws WR, Ross JBA, Shea MA (2002) Biophys J 83:2767–2780
51. Rossotti FJC, Rossotti H (eds) (1961) The determination of stability constants. Mc Graw-Hill Book Company Inc, New York
52. Scatchard G (1949) Ann NY Acad Sci 51:660–666
53. Hill AV (1913) Biochem J 7:471–480
54. Grabarek Z (2005) J Mol Biol 346:1351–1366
55. Brown A (2009) Int J Molecular Sciences 10:3457–3477
56. Brokx RD, Lopez MM, Vogel HJ, Makhatadze GI (2000) J Biol Chem 276:14083–14091
57. Crouch TH, Klee CB (1980) Biochemistry 19:3692–3698
58. Linse S, Chazin WJ (1995) Protein Sci 4:1038–1044
59. Busch E, Hohenester E, Timpl R, Paulsson M, Maurer P (2000) J Biol Chem 275:25508–25515
60. LeClainche L, Figuet M, Montjardet-Bas V, Blanchard S, Vita C (2006) Biotechnol Bioeng 95:29–36
61. Gilli R, Lafitte D, Lopez C, Kilhoffer MC, Makarov A, Briand C, Haiech J (1998) Biochemistry 37:5450–5456
62. Kilhoffer MC, Kubina M, Travers F, Haiech J (1992) Biochemistry 31:8098–8106

63. Haiech J, Klee CB, Demaille JG (1981) Biochemistry 20:3890–3894
64. Wu G, Gao Z, Dong A, Yu S (2012) Int J Biol Macromol 50:1011–1017
65. Dagher R, Peng S, Gioria S, Feve M, Zeniou M, Zimmermann M, Pigault C, Haiech J, Kilhoffer MC (2011) Bba-Mol Cell Res 1813:1059–1067
66. Henzl MT, Larson JD, Agah S (2003) Anal Biochem 319:216–233
67. Marcus Y (ed) (1985) Ion solvation. John Wiley, Chichester
68. Gifford JL, Walsh MP, Vogel HJ (2007) Biochemistry 405:199–221
69. Wimberly B, Thulin E, Chazin WJ (1995) Protein Sci 4:1045–1055
70. Katragadda M, Magotti P, Sfyroera G, Lambris JD (2006) J Med Chem 49:4616–4622
71. Kauzmann W (1959) Adv Protein Chem 14:1–63
72. Ma JC, Dougherty DA (1997) Chem Rev 97:1303–1324